Access-2026-38745 — English revision addendum

Integral article = 22-page English base PDF (SHA-256 CFD80F0D0EE9…) + these pages. Tables 1–20 keep their numbers. New material is Tables 21–27. Yellow marks new or replacement text. Not IEEE two-column Overleaf output. Collaborator pack; not the journal upload.

Deletes in the English base (V-B, V-D, VI-C, VII-A)

The English CFD80 PDF is the source. Portuguese evaluation pages from the older 5CBC file are not used. Delete four denials: (1) V-B, no Module32 positive control; (2) V-D, InstMemcmpMask restricted to memcmp; (3) VI-C, absence of monitored modules is not a positive control; (4) VII-A, absence of target modules prevents confirmation of Module32 filtering. Paste the replacements below.

V-B ProcessEnum — Module32 fixture (after Table 6)

The process oracle TestProcessEnum.exe used Unicode process enumeration and found three pin.exe entries under -q and none under -dp or -da (Table 6; one execution per configuration). A separate positive control exercised Module32 filtering (module32_fixture.csv). A copy of C:\Windows\System32\version.dll was loaded as %TEMP%\pinvm.dll. That file is a dummy name; it is not the real Pin module. Native and TMW0 enumerated the fixture. With -dp -q and -da -q the fixture name was hidden. kernel32.dll remained visible. Enumeration terminated (ENUM_OK). Counts: 32 / 31 / 30 / 30. The fixture does not validate NtQuerySystemInformation and does not individually validate every GetModuleHandleA/W calling pattern.

Configuration	Knobs	Exit	Modules	kernel32.dll	fixture pinvm.dll
Native	—	0	32	visible	visible
TMW0	-q	0	31	visible	visible
ProcessEnum	-dp -q	0	30	visible	hidden
Combined	-da -q	0	30	visible	hidden

TABLE 21. Module32 positive control. Dummy %TEMP%\pinvm.dll is a copy of version.dll, not a real Pin module. Source: module32_fixture.csv. Campaign DLL 4BBC6270.

Combined -da smoke test (not ten repetitions)

The filename da_n10.csv belongs to campaign id t2_n10_v7b. It contains five rows, one execution per oracle, not ten repetitions of combined mode. All five oracles exited 0. TestOverhead.exe scored 2042.56 < 3000. This smoke test does not prove the absence of module interactions on the 100-binary corpus. TMW_ALL in Table 19 is a separate operational arm.

Oracle	Knobs	Exit	Timed out	Duration (ms)
TestAntiDebug	-gdb -da -q	0	no	10316
TestProcessEnum	-da -q	0	no	13413
TestGetEnvironments	-da -q	0	no	13324
TestMemoryScan	-da -q	0	no	44062
TestOverhead	-da -q	0	no	14658

TABLE 22. Combined-mode smoke test (-da). One execution per oracle. Campaign DLL 4BBC6270.

InstMemcmpMask family (keep Table 12)

Keep Table 12 memcmp signature counts. On campaign DLL 4BBC6270, intercepted PIN-string comparisons were PASS under native and TMW0 and MASKED under -dm -q and -da -q for memcmp, _memicmp, and RtlCompareMemory. Legitimate equality and inequality (pass-through) remained OK. RtlEqualMemory, wmemcmp, memcmp_s, wmemcmp_s, and bcmp were ABSENT / NOT_EXERCISED. They are implemented in the Pintool but were not individually validated. _memicmp_l is FOUND in msvcrt. Native and TMW0: PASS. Under -dm and -da the PIN-string path printed MASKED and the pass-through aborted with RECEIVED_ACCESS_FAULT at 0xffffffffffffffff (exit 2; three-argument hook versus four-argument CRT). Later DLL D55EC952 does not retroactively validate the 9,000 attempts.

Configuration	Exit	memcmp PIN	_memicmp PIN	RtlCompareMemory PIN
Native	0	PASS (ret=0)	PASS (ret=0)	PASS (n=4)
TMW0 -q	0	PASS	PASS	PASS
-dm -q	2	MASKED (ret=1)	MASKED (ret=1)	MASKED (n=0)
-da -q	2	MASKED	MASKED	MASKED

TABLE 23. Comparison-family controls, campaign DLL 4BBC6270, memcmp_n10.csv, one execution per configuration.

Section VI — 22/30 diagnostic (not Introduction-only)

A separate diagnostic, not part of the 9,000-attempt corpus, compared TMW0 (-q) with TMW_ALL (-da) under an external observer (external_ps). The observer recorded the process exit code and the SHA-256 of standard output. It did not record API traces, code

coverage, or kernel events. Guest timeout 45 s. Ten executables (BEN_05, BEN_06, BEN_10, BEN_12, BEN_14, MAL_01, MAL_02, MAL_06, MAL_07, MAL_11) × three repetitions = 30 pairs. Agreement: equal exit code and equal stdout SHA-256. Result: 22 of 30 agreed; 8 disagreed. This is not API equivalence. Per-binary hook counters were not collected in the 9,000 JSON.

Pairs	TMW0	TMW_ALL	Reading
22/30	—	—	exit and stdout hash equal
BEN_06 × 3	exit 3	timeout (−2)	combined mode changed the outcome
BEN_12 × 2	exit 0, 754 B stdout	timeout; finished attempt same hash	text matched when the process exited
MAL_06 × 1	exit 0	exit 2	one of three repetitions differed
MAL_07 × 2	exit 0	0xC0000005	combined mode worsened termination

TABLE 24. External-observer diagnostic, TMW0 versus TMW_ALL, 10 executables × 3 repetitions. Source: fidelity_by_run.csv.

VI-B Baselines NAT / EMPTY / TMW0 and Pin without a Pintool

Two contrasts are separated. Total instrumentation cost is NAT versus EMPTY or TMW0. Incremental countermeasure effect is TMW0 versus the enabled module. TMW0 versus a module is not the total cost of Intel Pin. Short-lived benign binaries (n=25), median of per-binary medians from stratum_medians.csv: NAT 0.260 s, EMPTY 1.485 s, TMW0 1.973 s.

Condition	Meaning	Median wall-clock (s)
NAT	no Pin	0.260
EMPTY	Pin + EmptyPinTool	1.485
TMW0	Pin + TOMWare, knobs off	1.973

TABLE 26. Descriptive median wall-clock, short-lived benign binaries (n=25). Incremental effects remain Table 20.

An isolated probe of Pin without a Pintool (pin.exe -- TestAntiDebug.exe, 22 Oct 2026, pin_probe.csv) completed with exit code 0, empty stderr, and oracle stdout reporting no PEB anomaly. The CSV still records pin_condition_valid=False because of the campaign-day IPC/StartServer note. The 9,000-attempt campaign omitted the PIN arm (campaign.json). EMPTY remains the smallest Pin-hosted reference in those 9,000 attempts. Do not claim that Pin without a Pintool is always impossible on Pin 3.28.

Field	Value
command	pin.exe -- TestAntiDebug.exe
exit code	0
timed out	false
stderr	(empty)
stdout (excerpt)	oracle OK — no anomaly; PEB fields 0
CSV pin_condition_valid	False (campaign-day IPC/StartServer note)

TABLE 27. Isolated Pin-without-Pintool probe, 22 Oct 2026, 30 s timeout. Source: pin_probe.csv.

```
Ocorrências:
Resumo de ocorrências:
IsDebuggerPresent          : 0
CheckRemoteDebuggerPresent : 0
ProcessDebugPort           : 0
ProcessDebugObjectHandle   : 0
PEB.BeingDebugged          : 0
PEB.NtGlobalFlag           : 0
OK - nenhuma anomalia
------------------------------
stderr: (empty)
exit_code: 0
```

VI-B Corpus manifesto

The 100-row manifesto is experiment/corpus/corpus.csv. Columns: SHA-256, source, acquisition date 2026-10-14, class, coarse type, architecture x64, profile, packer/protector, targeted_evasion. Trojan_Bazaar and Backdoor_Bazaar are coarse MalwareBazaar labels, not VirusTotal families. Packer/protector remains unknown on the malware rows. targeted_evasion=unverified for all 50 malicious binaries and none for the 50 benign binaries.

VI-C — replacement of the Module32 denial

The five oracles assessed debugging, enumeration, the environment, memory comparisons, and timing. Those results are not evidence that corpus binaries performed the same checks. ProcessEnum includes a Module32 positive control on a dummy %TEMP%\pinvm.dll copy of version.dll (Table 21). Combined activation was smoke-tested once per oracle (Table 22).

InstMemcmpMask family controls cover memcmp, _memicmp, and RtlCompareMemory; _memicmp_l faulted on the campaign DLL; several comparison exports were not exercised (Table 23).

VI-D — 6,517 unavailable exit codes

Table 19 includes all 9,000 attempts. In the 9,000 JSON records, 6,517 rows were stored as exception_crash with a null exit_code and exception_code=0x. After Start-Process with redirected streams, Windows PowerShell 5.1 can leave Process.ExitCode unpopulated; a null compared with < 0 was treated as a crash. True crashes with a known exit code in those 9,000 records: 0. The original JSON files were not rewritten. Collector v2.1 (Get-SafeProcessExitCode) applies only to later campaigns. A success rate is not computed from the 6,517 rows.

Process-tree diagnostic access_r43b (BEN_28, BEN_46, BEN_47)

The TMW0 versus TMW_DO corpus contrast uses 2,000 attempts (1,000 per condition), not 9,000. An additional diagnostic (access_r43b_process_tree, script_rev=r43b.2) selected the same three persistent benign binaries a priori (campaign TMW0 n_ok=0, TMW_DO n_ok≥6): BEN_28, BEN_46, BEN_47. Native exited −1073741511 (STATUS_ENTRYPOINT_NOT_FOUND) with peak WS = 0 and no named tree. TMW0 BEN_28 and BEN_46 exited with the same code at 1625 ms; BEN_46 left a named tree pin.exe → sample + conhost.exe. TMW_DO BEN_46 remained STILL_ALIVE_AT_WINDOW at 61063 ms with that named tree. TMW_DO BEN_28 had a live three-node tree for about 49 s and then exited −1073741511. BEN_47 under TMW0 and TMW_DO is a Pin launch failure (Win32 740, elevation required), not a SkewMask hold. The earlier access_r43_skewmask_diag campaign is preserved; its process_tree.txt failed. identified=false. Persistence is state or wait, not application progress.

Sample	Mode	Exit	Elapsed (ms)	Progress
BEN_28	Native	−1073741511	3375	EXITED (WS = 0, no tree)
BEN_28	TMW0	−1073741511	1625	EXITED (n_tree=3 then dead)
BEN_28	TMW_DO	−1073741511	49344	EXITED after live n_tree=3
BEN_46	Native	−1073741511	1155	EXITED (WS = 0, no tree)
BEN_46	TMW0	−1073741511	1625	EXITED; pin.exe→sample+conhost
BEN_46	TMW_DO	−1 (timeout)	61063	STILL_ALIVE; same named tree
BEN_47	Native	−1073741511	437	EXITED (WS = 0, no tree)
BEN_47	TMW0	−1	719	Pin fatal 740 (elevation)
BEN_47	TMW_DO	−1	453	Pin fatal 740 (elevation)

TABLE 25. Process-tree diagnostic access_r43b_process_tree, N=1, 60 s, DLL 4BBC6270. identified=false. BEN_47 TMW0/TMW_DO is Pin Win32 740, not STATUS_ENTRYPOINT_NOT_FOUND.

VI-E Resource pairs (already in the English Methods)

The resource-contrast filter is post hoc and is not in experiment.config.json. Duration: 98 eligible pairs. CPU and memory: 98 pairs, except SanitizePinEnvVars, which had 97 (BEN_32). MAL_33 and MAL_37 remain in the 9,000 outcome tables. Intervals are exploratory.

VII-A — replacement of the Module32 denial

ProcessEnum hid the pin.exe process entries on the Tool Help path that the process oracle exercised, and hid the dummy pinvm.dll fixture on the Module32 path (Table 21). The fixture is not the real Pin module. GetModuleHandleA/W and native NtQuerySystemInformation remain without individual validation.

Remaining experimental limitations (not marked resolved)

No new experiments were run for: (1) API traces and code coverage; (2) per-binary hook-invocation counters (disabled in the 9,000); (3) full temporal coherence; (4) a malware case with identified check, branch, matching knob, and recovered hashed behavior; (5) experimental comparison with Arancino, COBAI, or Peekaboo.